INCH-POUND

MIL-V-52830B(ME)
22 September 1992
SUPERSEDING
MIL-V-52830A(ME)
22 May 1980

MILITARY SPECIFICATION

VIEWING STAND, ILLUMINATED, NEGATIVE FILM

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an illuminated viewing stand with a 24- by 30-inch viewing area.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: SATBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6760

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SPECIFICATIONS

FEDERAL

J-C-580	- Cord, Electrical and Wire, Electrical (0- to 600-Volt Service).
W-C-596	- Connector, Electrical Power.
W-J-800	- Junction Box; Extension, Junction Box, Cover, Junction Box (Steel, Cadmium, or Zinc-Coated).
W-L-101	- Lamp, Incandescent, (Electric, Large, Tungsten-Filament) (General Specification).
W-L-116	- Lamp, Electric Incandescent Miniature, Tungsten Filament.
W-L-142	- Lampholder, Adaptor, and Shadeholder, Medium-Screw-Shell, 125, 250 and 600 Volts.
QQ-A-250	- Aluminum and Aluminum Alloy Plate and Sheet, General Specifications for.
QQ-A-250/8	- Aluminum Alloy 5052, Plate and Sheet.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3432	- Cable, (Power and Special Purposes) and Wire, Electrical (300 and 600 Volts).
MIL-T-7928	- Terminal, Lug Splices, Conductor, Crimp Style, Copper, General Specifications for.
MIL-W-5086	- Wire, Electric, Polyvinyl Chloride Insulated, Copper or Copper Alloy.
MIL-P-24191	- Plastic Sheet, Cast, Acrylic, Shipboard Application (Illumination and Signal Lighting).
MIL-R-24243	- Rivet, Blind, Nonstructural, Retained Mandrel, General Specification for.

STANDARDS

FEDERAL

FED-STD-595	- Color in Government Procurement.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.

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MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.
MS-3367	- Strap, Tiedown, Electrical Components, Adjustable, Self Clinching, Plastic, Type 1, Class 1.
MS16206	- Bolt, Machine, Hexagon Head, Regular Semi-Finished, Aluminum Alloy, UNC-2A, Nonmagnetic.
MS25036	- Terminal, Lug, Crimp Style, Copper, Insulated, Ring-Tongue, Bell Mouthed, Type II, Class 1. (For 105 Deg. C Total Conductor Temperature).
MS25274	- Cap, Electrical (Wire End, Crimp Style, Type II, Class 1) For 105 Deg. C Total.
MS25307	- Switch, Toggle, Positive Break, Environmentally Sealed Screw Terminal, Double Pole, .469 Mounting Bushing 25 Amperes.
MS15795	- Washer, Flat Metal, Round, General Purpose.
MS51481	- Screw, Machine, 82 Degree Flat Countersunk Head, Cross-Recessed, Carbon Steel, UNC-2A.
MS51957	- Screw, Machine-Pan Head, Cross-Recessed, Corrosion-Resistant Steel UNC-2A.
MS35338	- Washer, Lock-Spring, Helical, Regular (Machine) Series.
MS35489	- Grommet, Synthetic and Silicone Rubber, Hot Oil and Coolant Resistant.
MS35649	- Nut, Plain, Hexagon, Machine Screw UNC-2B.
MS35825	- Hinge, Butt, Continuous (Piano) .060 Thickness .125 Pin Dia.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DRAWINGS

ME

D13214E8326	- Welding Requirement for Aluminum and Aluminum Alloy.

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: SATBE-JD, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of

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the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A684	- Steel, Carbon, Strip, Cold-Rolled Soft, Untempered Spring Quality.
D3951	- Standard Practice for Commercial Packaging.
D3953	- Strapping, Flat Steel and Seals.
D4675	- Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103).

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The viewing stand shall be as shown on figures 1, 2 and 3, and as specified herein.

3.2 Material. Material shall be as specified herein and as shown on the applicable figures. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.2.1 Material deterioration prevention and control. The viewing stand shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the viewing stand may be exposed.

3.2.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.2.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

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3.2.4 Aluminum. Aluminum sheet shall conform to QQ-A-250 and QQ-A-250/8, Temper H32.

3.2.5 Steel. Steel spring shall conform to ASTM A 684, grade optional, finish optional.

3.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the viewing stand may be newly fabricated from recovered materials to the maximum extent practicable, provided the item produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the viewing stand.

3.4 Plastic window. The window shall be of plastic conforming to MIL-P-24191, type WTLS-4, grade D.

3.5 Incandescent lamps. Incandescent lamps shall conform to W-L-101, bulb S-14. The bulb shall be dyed red to produce a red light.

3.5.1 Lampholder. The lampholder shall conform to W-L-142 and be of the switchless type and shall have a porcelain base to fit the junction box specified in 3.6. The lampholder shall have a screw-base socket to fit the lamp specified in 3.5.

3.6 Junction box. The junction box shall conform to W-J-800, type 1, size B, style 7.

3.7 Fluorescent lamps. Fluorescent lamps shall conform to W-L-116, 20-watt, T-12 bulb, daylight, 24-inch.

3.8 Switch. The switch shall conform to MS25307 and be of the Double-Pole-Double-Throw (DPDT) type with center "OFF" position. The switch shall be rated at not less than 120-volt, 20-ampere. The switch shall be wired as shown on the wiring diagram on the figures so that one position will turn on the incandescent lamps and the other position will turn on the fluorescent lamps. The switch shall be labeled "Light" above the switch in 0.25 inch high letters. The positions of the switch shall be labeled "Off" in the center position, "Red" for the incandescent light position, and "White" in the fluorescent light position in 0.18 inch high letters.

3.9 Wiring. Interior wiring shall conform with MIL-C-3432, MIL-W-5086, MS25036, and MS25274. Wiring shall be routed along the inside walls of the viewing stand. Wiring shall be supported to prevent sagging, and grommets shall conform to MS35489 and be provided at all points where wiring passes through holes. Electrical component tiedown straps must conform to MS3367.

3.9.1 Cord. A cord not less than 10 feet long shall be connected to the stand to provide power for the lamps. The cord shall conform to J-C-580, type

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and size optional, 2-wire with ground. The cord shall terminate with a plug conforming to W-C-596, 140-1.

3.10 Human factors engineering. The viewing stand shall comply with the design criteria of MIL-STD-1472. Special design emphasis shall be given, but not limited to, MIL-STD-1472, general requirement (4), controls (5.4), labeling (5.5), design for maintainer (5.9), small systems and equipment (5.11), and hazards and safety (5.13), as applicable.

3.11 Identification marking. The light stand shall be identified in accordance with MIL-STD-130.

3.12 Treatment and painting. Treatment and painting shall be in accordance with MIL-T-704, type A, except that the interior shall be gloss white and the exterior shall match color 34088 of FED-STD-595. When specified (see 6.2), the exterior shall match No. 24525 or 24533 of FED-STD-595.

3.12.1 Hardware. All screws, nuts, bolts, washers, and lock washers shall be zinc-plated and conform to MS15795, MS51481, MS51957, MS35338, M16206 and MS35649 where applicable. Continuous butt hinges shall conform to MS35825. Cadmium plating/plated item shall not be used.

3.13 Workmanship.

3.13.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of materials shall be done by methods that will not cause injury to the material. Corners shall be square and true. All bends shall be made with controlled means to ensure uniformity of size and shape. Surfaces shall be free from burrs, sharp edges, and sharp corners.

3.13.2 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers and lock washers shall be provided where necessary. All fasteners shall be correctly torqued. All bolts shall have full thread engagement, and bolts shall not extend more than two full threads.

3.13.3 Riveted connections. Rivets shall conform to MIL-R-24243 and fill the hole completely. The upset rivet head shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.13.4 Welding. Welding shall be in accordance with 13214E8326, type III, class 2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or

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any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspection shall be classified as follows:

a. Quality conformance inspection (see 4.3).
b. Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sample size shall be determined by table I and table IIa of MIL-STD-105. A lot shall be accepted when 0 defects are found, and rejected when one or more defects are found.

4.4 Inspection procedure.

4.4.1 Examination. Samples selected in accordance with 4.3.1 shall be examined for the following defects.

101. Viewing stand not in accordance with the figures (see 3.1).
102. Materials not as specified (see 3.2).
103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.2.1).
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.2.2).
105. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.23).
106. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the viewing stand (see 3.3).
107. Plastic window not as specified (see 3.4).

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108. Incandescent lamps not dyed red, not of the type specified (see 3.5).
109. Junction box not as specified (see 3.6).
110. Fluorescent lamps not as specified (see 3.7).
111. Switch not as specified; does not function as specified (see 3.8).
112. Wiring not as specified (see 3.9).
113. Human factors engineering not as specified (see 3.10).
114. Identification marking missing, incomplete or illegible (see 3.11).
115. Treatment and painting not as specified (see 3.13).
116. Cadmium plating/plated parts used (see 3.12.1).
117. Workmanship not as specified (see 3.13).

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sample size shall be determined by table I and table IIa of MIL-STD-105. A lot shall be accepted when 0 defects are found and rejected when one or more defects are found.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

118. Preservation not as specified for level A (see 5.1.1).
119. Methods or containers not as specified for level A and B (see 5.1.1, 5.2.1, and 5.2.2).
120. Boxes not closed in accordance with method V as specified (see 5.1.1).
121. Preservation or packing for commercial not as specified in accordance with the referenced document (see 5.1.2 and 5.2.3).
122. Marking illegible, incorrect, incomplete, or missing (see 5.3).

5. PACKING

5.1 Preservation. Preservation shall be level A, or commercial, as specified (see 6.2).

5.1.1 Level A. Each viewing stand shall be preserved in accordance with MIL-P-116, method III, in a close-fitting box conforming to PPP-B-636, W6c, style optional. Closure of the box shall be in accordance with the appendix to the box specification, method V.

5.1.2 Commercial. Each viewing stand shall be preserved in accordance with ASTM D 3951.

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5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Viewing stands preserved as specified in 5.1 shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style optional, grade B. The box closure and strapping shall be in accordance with ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

5.2.2 Level B. Viewing stands shall be packed as specified for level A, except that boxes shall be domestic type and the strapping may be zinc-coated.

5.2.3 Commercial. Viewing stands, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 MARKING

5.3.1 Military. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The viewing stand is intended for viewing and inspecting negatives and other photographic material used in the lithographic process, and is used in components list 3610-97-CL-E13.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When specified exterior color is required (see 3.12).
d. Degree of preservation and level of packing required (see 5.1 and 5.2).
e. Any special marking required (see 5.3).

6.3 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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6.4 Subject term key word listing.

Mapping
Inspecting negatives
Photographic material
Lithographic process

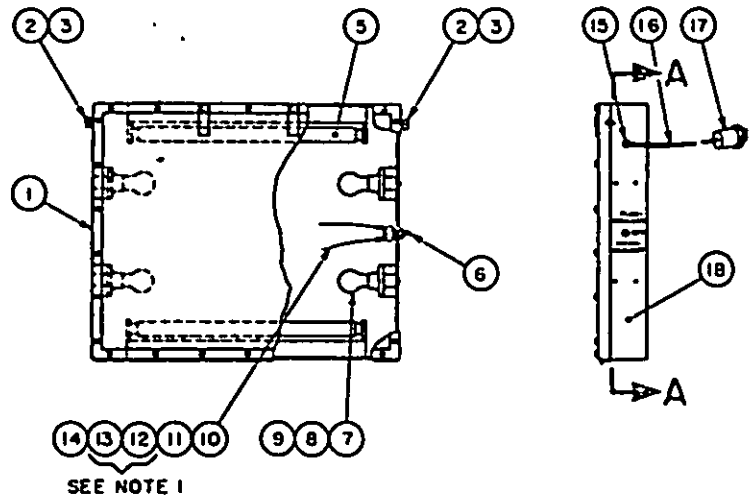
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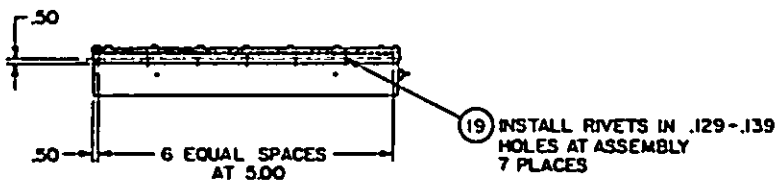
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NOTES:

1. INSTALLATION OF TERMINALS, FIND NO. 12, AND CAPS, FIND NO. 13, SHALL MEET THE PERFORMANCE REQUIREMENTS OF MIL-T-7928.
2. SUGGESTED SOURCE OF SUPPLY, REID TOOL SUPPLY CO., 2233 TEMPLE ST. MUSKEGON HEIGHTS MI, FSCM 01599, PART NO. SST-2T



LIST OF MATERIAL

FIND NO.	QTY	DESCRIPTION	MATERIAL OR PART NO.
1	1	LENS ASSEMBLY .063 STK	AL ALY
2	2	KNOB, ALUM, 1/4-20 UNC	SEE NOTE 2
3	2	WASHER, FLAT, 1/4	MS15795
5	2	LIGHT, FLUORESCENT, T-12, MEDIUM BIPIN	W-L-116
6	1	SWITCH, TOGGLE	MS25307-212
7	4	JUNCTION BOX	W-J-800, TYPE I, SIZE A, STYLE 2 OR 4
8	4	LAMPHOLDER, OUTLET BOX MTG	W-L-142, TYPE V-25
9	4	LIGHT, INCANDESCENT, 10W, 120V, S-14 RED	W-L-101/98
10	AR	WIRE, NO. 16 WHITE	MIL-W-5086/1 M5086/1-16-9
11	AR	WIRE, NO. 16 BLACK	MIL-W-5086/1 M5086/1-16-0
12	16	TERMINAL LUG	MS25036-107
13	AR	CAP, ELECTRICAL	MS25274-3
14	AR	STRAP, TIEDOWN ELECTRICAL	MS3367-1-9
15	1	GROMMET	MS35489-11
16	AR	CABLE, BLK, GRN, WH COND.	MIL-C-3432 CO-03 MGF (3/16) 0365
17	1	PLUG	W-C-596/14C
18	1	BOX, LIGHT .063 STK	AL ALY
19	7	RIVET, BLIND, .125 Ø X .063-.125 GRIP	MIL-R-24243 M24243/6-A402H
20	12	SCREW, PAN HD, .164-32 UNC-2A X .50 L	MS51957
21	12	WASHER, FLAT, NO. 8	MS15795
22	12	WASHER, LOCK, NO. 8	MS35338-42
23	12	NUT, HEX .164-32 UNC-2B	MS35649-282
24	2	STIFFENER, FILM CLIP, .030 STK	SPRING STL
25	2	CLIP, FILM .030 STK	SPRING STL
26	26	SCREW, FLAT HD, .138-32 UNC-2A X .50 L	MS51481
27	26	NUT, HEX .138-32 UNC-2B	MS35649-262
28	1	DIFFUSER .25 STK	PLASTIC SHEET
29	1	HINGE, .06 STK, .125 Ø PIN, 31.00 L	MS35825-11
30	2	BOLT, HEX HD, 1/4-20 UNC-2A X .75 L	MS16206-5

FIGURE 1. Light box general assembly and details.

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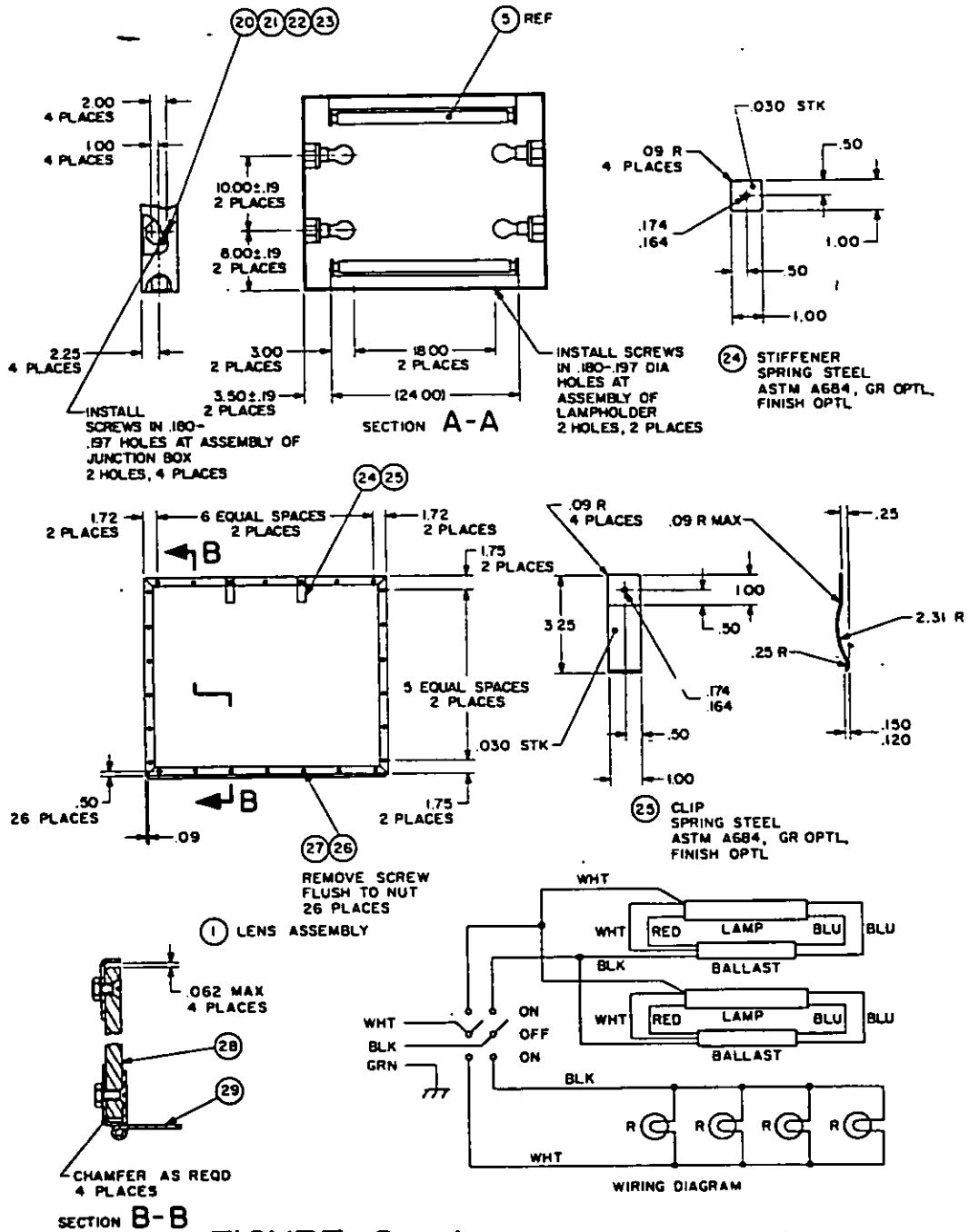


FIGURE 2. Light box general assembly and details.

X-3660

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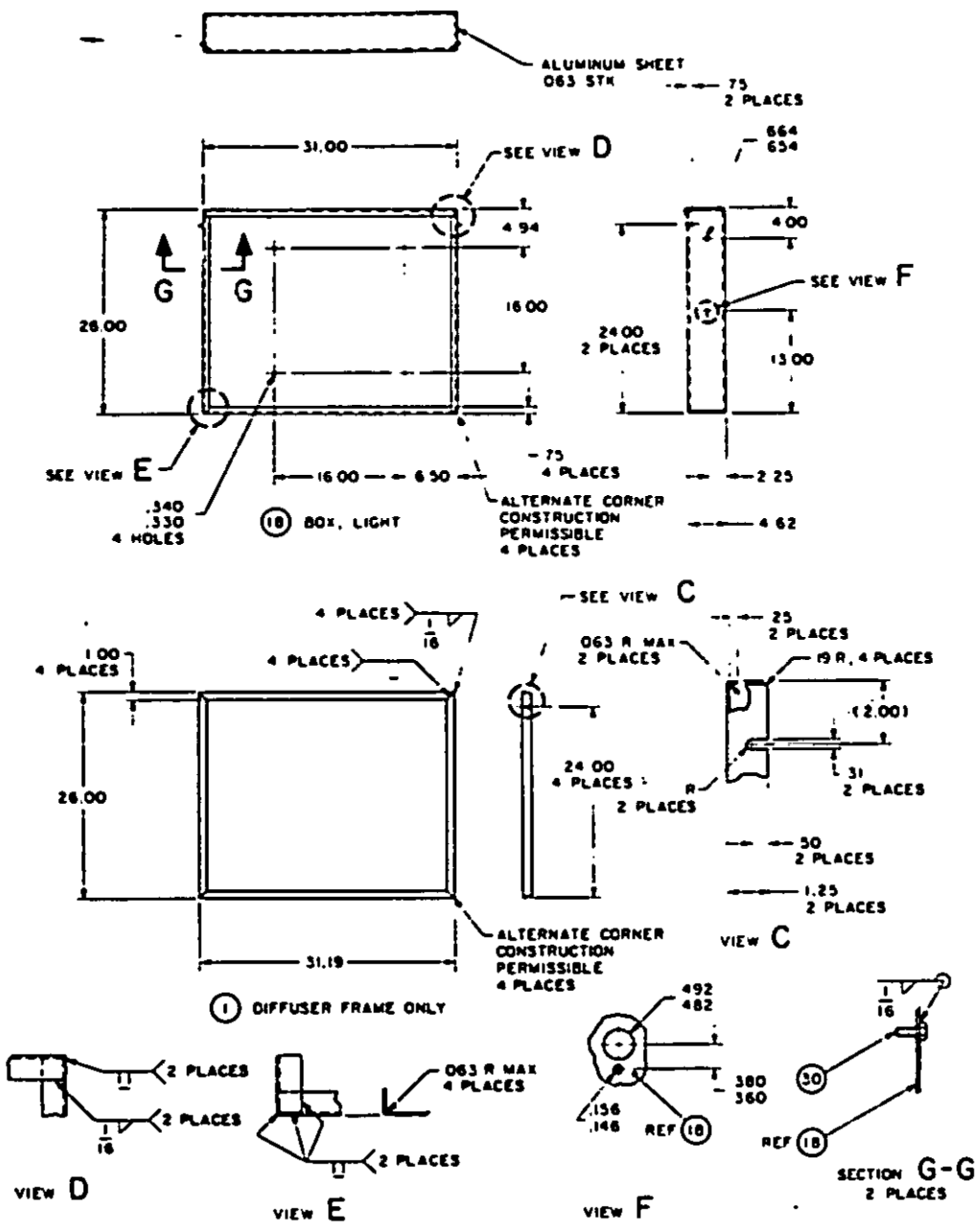


FIGURE 3. Viewing stand: details.

X-2736A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-V-52830B(ME)	2. DOCUMENT DATE (YYMMDD) 920922

3. DOCUMENT TITLE
Viewing Stand, Illuminated, Negative Film

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Violet Stewart	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3467	(2) AUTOVON 654-3467
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: SATBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340	AUTOVON 289-2340

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